DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III **Topics in Analysis (3TN)**

Algebraic operations on Fréchet differentiable mappings

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In these lectures, we want to extend the usual addition and product rules as well as the Chain rule for the derivative (known in our basic calculus courses) to the present context of the Fréchet differentiability of mappings beween normed spaces.

1 Sum and scalar mulplication

Proposition 1.1 *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two Banach spaces over the field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let Ω be an open subset of V. Let $f, g : \Omega \to W$ and $\lambda : \Omega \to \mathbb{K}$ be Fréchet differentiable at x. Then the functions $(f+g)$ and (λf) are also Fréchet differentiable at x and for any $v \in V$*

$$\begin{cases} D(f+g)(x)(v) = Df(x)(v) + Dg(x)(v) \text{ and} \\ \\ D(\lambda f)(x)(v) = \lambda(x) Df(x)(v) + (D\lambda(x))(v) f(x)\,. \end{cases}$$

Proof: Notice that

$$\begin{aligned} &\lim_{h\in V,\, \|h\|_V} \frac{(f+g)(x+h) + (f+g)(x) - Df(x)(h) - Dg(x)(h)}{\|h\|_V} \\ = &\lim_{h\in V,\, \|h\|_V} \frac{f(x+h) + f(x) - Df(x)(h)}{\|h\|_V} + \lim_{h\in V,\, \|h\|_V} \frac{g(x+h) + g(x) - Dg(x)(h)}{\|h\|_V} = 0 \end{aligned}$$

and $Df(x)+Dg(x) \in \mathcal{L}(V, W)$. Hence, $(f+g)$ is Fréchet differentiable at x and D(f+g)(x)=Df(x)+Dg(x). Now

$$\begin{aligned} &\lim_{h\in V,\, \|h\|_V\to 0} \frac{(\lambda f)(x+h) + (\lambda f)(x) - \lambda(x)Df(x)(h) - D\lambda(x)(h)f(x)}{\|h\|_V} \\ = \;& \lambda(x) \lim_{h\in V,\, \|h\|_V\to 0} \frac{f(x+h) - f(x) - Df(x)h}{\|h\|_V} + \lim_{h\in V,\, \|h\|_V\to 0} \frac{\lambda(x+h) - \lambda(x) - D\lambda(x)(h)}{\|h\|_V} f(x+h) \\ &+ \lim_{h\in V,\, \|h\|_V\to 0} \frac{D\lambda(x)(h)}{\|h\|_V}[f(x+h) - f(x)] = \lim_{h\in V,\, \|h\|_V\to 0} \frac{D\lambda(x)(h)}{\|h\|_V}[f(x+h) - f(x)]\,. \end{aligned} \tag{1}$$

Now, notice that $D\lambda(x) \in \mathcal{L}(V, \mathbb{K})$ and f is continuous at x Thus

$$|D\lambda(x)(h)| \le \|D\lambda(x)\|_{\mathcal{L}(V,\mathbb{K})} \|h\|_V \quad \text{ and } \lim_{h\in V,\, \|h\|_V\to 0} \|f(x+h) - f(x)\|_W = 0\,.$$

Therefore

$$\lim_{h\in V,\, \|h\|_V\to 0}\left\|\frac{D\lambda(x)(h)}{\|h\|_V}[f(x+h)-f(x)]\right\|_W \le \|D\lambda(x)\|_{\mathcal{L}(V,\mathbb{K})}\left[\lim_{h\in V,\, \|h\|_V\to 0}\|f(x+h)-f(x)\|_W\right] = 0\,.$$

Hence, from (1) we get that

$$\lim_{h\in V,\, \|h\|_V\to 0}\frac{(\lambda f)(x+h)+(\lambda f)(x)-\lambda(x)Df(x)(h)-D\lambda(x)(h)f(x)}{\|h\|_V} = 0\,.$$

On the other, the mapping $v\in V\to \lambda(x)Df(x)(v)-D\lambda(x)(v)f(x)\in W$ is linear and continuous. Therefore, (λf) is Fréchet differentiable at x and for any $v\in V$

$$D(\lambda f)(x)(v) = \lambda(x)Df(x)(v) + (D\lambda(x))(v)f(x)\,. \quad \blacksquare$$

Remark 1.2 In the case $X=\mathbb{R}^n$ and $Y=\mathbb{R}$, we define the partial derivatives of f at $x=(x_1,\dots,x_n)$ as the directional derivatives of at x along the directions $e_i=(0,\dots,1,\dots,0)$ for any $i=1,2,\dots,n$. That is, the partial derivative of f with respect to x_i is defined as

$$\frac{\partial f}{\partial x_i}(x) = \lim_{h\to 0}\frac{f(x+he_i)-f(x)}{h} \qquad \text{provided the limit exists.}$$

The following notations are also used: f_{x_i} or $D_i f$.
From the fact that the Fréchet differentiability implies the Gâteaux differentiability (as proved in the Lecture L11, it follows that, if $f:\Omega\subset R^n\to\mathbb{R}$ is differentiable at x, then for any $v=\sum_{i=1}^n v_i e_i$ we have

$$f_v(x) = Df(x)(v) = \sum_{i=1}^n v_i Df(x)(e_i) = \sum_{i=1}^n v_i f_{x_i}(x)\,.$$

The vector $\nabla f(x) = (f_{x_1}(x),\dots,f_{x_n}(x))$ is called the *gradient of f* at x.

A wel- known sufficient condition for the Fréchet differentiability of a function $f:\Omega\subset\mathbb{R}^n\to\mathbb{R}$ is given in the next theorem. The following theorem, was proved last year in the module Advanced Calculus (2AC)

Theorem 1.3 *Let $f:\Omega\subset\mathbb{R}^n\to\mathbb{R}$ be function having partial derivatives at any point in an open neighborhood $U\subset\Omega$ of x. If for every $i\in\{1,\dots,n\}$, f_{x_i} is continuous at x, then f is differentiable at x.*

Proof: We consider the simple case $n=2$. That is, f is a function of two variables $x,\, y\in\Omega\subset\mathbb{R}^2$. We write

$$f(x+h,y+k)-f(x,y) = [f(x+h,y+k)-f(x,y+k)] + [f(x,y+k)-f(x,y)]\,.$$

Now we apply the mean value theorem twice and get ξ between x and $x+k$ and η between y and $y+k$ such that

$$f(x+h,y+k)-f(x,y+k) = f_x(\xi,y+k)h + f_y(x,\eta)k\,.$$

Hence,

$$\left|\frac{f(x+h,y+k)-f(x,y)-f_x(x,y)h-f_y(x,y)k}{\sqrt{h^2+k^2}}\right| \le |f_x(\xi,y+k)-f_x(x,y)|\cdot\frac{|h|}{\sqrt{h^2+k^2}}+$$
$$+|f_y(x,\eta)-f_y(x,y)|\cdot\frac{|k|}{\sqrt{h^2+k^2}} \le |f_x(\xi,y+k)-f_x(x,y)|+|f_y(x,\eta)-f_y(x,y)|.$$

Now as $(h,k)\to(0,0)$ we have that $(\xi,\eta)\to(x,y)$. Now using the continuity of f_x and f_y at (x,y), it follows

$$f_x(\xi,y+k)-f_x(x,y)\to 0 \quad \text{and} \quad f_y(x,\eta)-f_y(x,y)\to 0\,.$$

Therefore

$$\lim_{(h,k)\to(0,0)}\frac{f(x+h,y+k)-f(x,y)-f_x(x,y)h-f_y(x,y)k}{\sqrt{h^2+k^2}}=0\,.$$

Hence, f is Fréchet differentiable at (x,y). $\square$

Definition 1.4 Let $(V_1,\|\cdot\|_1)$ and $(V_2,\|\cdot\|_2)$ be two normed spaces and $\Omega\subset V_1$ be an open set. We say that the function $f:\Omega\to V_2$ is of class $\mathcal{C}^1$ if f is differentiable and the mapping $Df:\Omega\to\mathcal{L}(V_1,V_2)$ is continuous. We set

$$\boxed{\mathcal{C}^1(\Omega,V_2):=\{f:\Omega\to V_2 \text{ of class } C^1\} \quad \text{and} \quad \mathcal{C}^1(\Omega):=\mathcal{C}^1(\Omega,\mathbb{R})}\,.$$

Let $k\ge 2$ be an integer. We say that $f:\Omega\to V_2$ is of class $\mathcal{C}^k$ if f is differentiable and the mapping $Df:\Omega\to\mathcal{L}(V_1,V_2)$ is of class $\mathcal{C}^{k-1}$. That is,

$$\boxed{\mathcal{C}^k(\Omega,V_2):=\{f:\Omega\to V_2 \text{ differentiable such that } Df\in\mathcal{C}^{k-1}(\Omega,\mathcal{L}(V_1,V_2)\}\,.}$$

2 The Chain Rule

The chain rule is about the Fréchet differentiability of a composition of fréchet differentiable functions.

Theorem 2.1 (Chain Rule)
Let $(V_1,\|\cdot\|_1)$, $(V_2,\|\cdot\|_2)$ and $(V_3,\|\cdot\|_3)$ be three normed spaces. Let Ω_1 be an open subset of V_1 and Ω_2 be an open subset of V_2. Let $f_1:\Omega_1\to V_2$ and $f_2:\Omega_2\to V_3$ be two functions such that $f(\Omega_1)\subset\Omega_2$. If f_1 is Fréchet differentiable at x and f_2 is Fréchet differentiable at $f_1(x)$. Then $f_2\circ f_1$ is Fréchet differentiable at x and

$$\boxed{D(f_2\circ f_1)(x)=Df_2(f_1(x))\circ Df_1(x)}$$

where the right hand side is the composition of the linear continuous maps $Df_2(f_1(x))\in\mathcal{L}(V_2,V_3)$ and $Df_1(x)\in\mathcal{L}(V_1,V_2)$.

Proof: The composition of linear continuous maps is linear continuous. Therefore the mapping $Df_2(f_1(x)) \circ Df_1(x) : V_1 \to V_3$ is linear continuous. That is, $Df_2(f_1(x)) \circ Df_1(x) \in \mathcal{L}(V_1, V_3)$. Now, from the previous lecture we write

$$\forall y \in \Omega_1, \quad f_1(y) = f_1(x) + Df_1(x)(y-x) + \|y-x\|_1 \, \omega_1(y)$$

and

$$\forall z \in \Omega_2. \quad f_2(z) = f_2(f_1(x)) + Df_2(f_1(x))(z - f_1(x)) + \|z - f_1(x)\|_2 \, \omega_2(z)\,.$$

Now taking $z = f_1(y)$ in the second equation we get that

$$\begin{aligned}
(f_2 \circ f_1)(y) - (f_2 \circ f_1)(x) &= Df_2(f_1(x))(f_1(y) - f_1(x)) + \|f_1(y) - f_1(x)\|_2 \, \omega_2(f_1(y)) \\
&= Df_2(f_1(x))(Df_1(x)(y-x)) + \|y-x\|_1 \, Df_2(f_1(x))(\omega_1(y)) \\
&+ \|Df_1(x)(y-x) + \|y-x\|_1 \, \omega_1(y)\|_2 \, \omega_2(f_1(y))
\end{aligned}$$

Thus it follows that

$$\begin{aligned}
&\frac{\|(f_2 \circ f_1)(y) - (f_2 \circ f_1)(x) - (Df_2(f_1(x)) \circ Df_1(x))(y-x)\|_3}{\|y-x\|_1} \leq \|Df_2(f_1(x))(\omega_1(y))\|_3 \\
&\quad + \frac{\|Df_1(x)(y-x)\|_2}{\|y-x\|_1} \|\omega_2(f_1(y))\|_3 + \|\omega_1(y)\|_2 \, \|\omega_2(f_1(y))\|_3 \\
&\leq \|Df_2(f_1(x))(\omega_1(y))\|_3 + \|Df_1(x)\|_{\mathcal{L}(V_1,V_2)} \, \|\omega_2(f_1(y))\|_3 + \|\omega_1(y)\|_2 \, \|\omega_2(f_1(y))\|_3
\end{aligned}$$

Now since f_1 is differentiable at x, f_1 is continuous at x and hence,

$$\lim_{y \to x} \omega_1(y) = 0, \quad \lim_{y \to x} \omega_2(f_1(y)) = 0, \quad \text{and} \quad \lim_{y \to x} Df_2(f_1(x))(\omega_1(y) = 0\,.$$

Thus

$$\lim_{y \to x} \frac{\|(f_2 \circ f_1)(y) - (f_2 \circ f_1)(x) - (Df_2(f_1(x)) \circ Df_1(x))(y-x)\|_3}{\|y-x\|_1} = 0\,.$$

Hence, $f_2 \circ f_1$ is Fréchet differentiable at x and

$$D(f_2 \circ f_1)(x) = Df_2(f_1(x)) \circ Df_1(x)\,. \quad \blacksquare$$

The next lecture will be on the Lagrange inequality, which can replace the identity in the mean value theorem (which does not always hold in the case of functions taking values in vector spaces of dimension greater than or equal to two).